Northern University of Business and Technology, Khulna

Department of Computer Science and Engineering

CSE 4104 – Software Development III

INITIAL PROJECT PLAN

Khabar Bondhu (KindMeal)

Team CSE4104-7B-T02

Section 7B

1. About This Plan

This plan shows how our team will build Khabar Bondhu over the 14 weeks of the course. I feel excited about this plan. We follow the phases set in the course handbook but we have filled each week with the actual features and checks for our own project. Each week also has a "done when" line so we can tell honestly

whether we are, on track. We have not written calendar dates because the dates depend on when each task's posted in Google Classroom. We will add them as the deadlines are announced.

2. Who Does What

Our team has four members, so each person leads one area. Every member also writes and tests code.

Role	What this person looks after
Team Leader	Keeps the weekly plan, divides tasks, checks that everyone's work is merged on time, talks to the instructor, and makes the Classroom submissions.
Frontend Developer	The React screens, forms and dashboards, the Need Map, the chatbot screen, and making everything work on phones and laptops.
Backend Developer	The Node.js and Express APIs, login and roles, the PostgreSQL tables, and checking the data that users send.
AI Integration Lead	The Python FastAPI service: reading expiry dates with OCR, matching donations to institutions, the Bangla chatbot, and testing the prompts.

3. What We Build First

A semester is short, so we have decided the order in advance. If time runs out, we cut from the bottom of this list, not from the top.

Priority	Items
Must have	Registration and login with roles; institution profiles with verification levels; food listing and pickup scheduling; donation requests with status tracking; admin dashboard; OCR expiry-date check with safety flag; Bangla chatbot; donor-to-institution matching; deployment.
Should have	Need Map with general areas only; in-app notifications; expense records and completion reports for donations.
If time allows	Email notifications; extra chatbot topics; more polished charts on the admin dashboard.

Priority	Items
Future work	Route planning, fraud detection, demand prediction, donor recommendations, AI impact analysis.

4. Week-by-Week Plan

Week	Focus	What we will do	Lead	Done when
1	Team and idea	Form the team, choose the idea, study similar systems.	All	Week 01 document submitted. (Completed)
2	Proposal and repository	Write the proposal and team sheet, create the GitHub repository, add everyone, upload the starter structure and README.	Team Leader	All four members can push to the repository.
3	SRS	Write requirements for each module, list the five user roles, describe the main use cases, draw a first use case diagram.	All	Every planned feature has at least one written requirement we can test.
4	System design	Draw the architecture (React, Express, PostgreSQL, FastAPI), the ER diagram (users, institutions, listings, pickups, donations, expenses) and the API list.	Backend	All members agree on the tables and endpoints before coding starts.
5	UI/UX	Design Figma screens: login, donor dashboard, institution profile, food listing form, Need Map, chatbot, admin verification.	Frontend	Each main screen is designed and reviewed by the team.

Week	Focus	What we will do	Lead	Done when
6	Backend	Set up the database; build registration, login (hashed passwords, tokens), roles, and APIs for institutions, listings, pickups and donations.	Backend	Every endpoint works in Postman with both correct and wrong input.
7	Frontend	Build the pages, connect them to the APIs, add loading and error messages, make the layout responsive.	Frontend	A user can register, log in, list food and view institutions from a browser on a laptop and a phone.
8	AI integration	Build the FastAPI service. Add OCR for expiry dates, the matching logic, and the Bangla chatbot. Try several prompts and keep the best.	AI Lead	OCR reads dates from our sample packages, matching returns a ranked list, and the chatbot answers sample Bangla questions.
9	Finish and catch up	Finish leftover features, add the Need Map and notifications, fix known bugs. This week is also our buffer for anything that slipped.	All	The feature checklist is complete and nothing important is left half-built.
10	Testing	Test features, APIs, screens on different devices, and the AI parts with many sample inputs. Record each bug and its fix.	All	Test report and bug-fix report are written and the major bugs are fixed.

Week	Focus	What we will do	Lead	Done when
11	Deployment	Deploy the frontend, backend, database and AI service. Set environment variables on the hosting services.	Backend	The live link works from another device and no keys appear in the repository.
12	Documentation	Write the final report, the user manual and the API documentation.	Team Leader	Each document is reviewed by someone who did not write it.
13	Presentation	Prepare slides, practise the demo, record a backup video and take backup screenshots.	All	We can run the whole demo twice without help.
14	Final	Present and answer viva questions.	All	Each member can explain their own part and the full workflow.

5. How We Will Work Together

- **One repository:** cse4104-7b-t02-khabar-bondhu on GitHub. All four of us are collaborators.
- **Branches:** main for stable work, development for putting things together, and separate branches for frontend, backend and ai-integration. We merge into development first and move to main only after testing.
- **Commits:** we commit at least weekly, in small steps, with clear messages such as “Added JWT authentication”. We avoid messages like “update” or “final”.
- **Code review:** before a branch is merged, another member looks at it. This also helps each of us understand the others’ code before the viva.
- **Weekly meeting:** one short meeting each week to see what was finished, what is blocked, and who does what next. A shared task board keeps track of the tasks.

- **Folders:** frontend, backend, ai-service, documentation, screenshots, testing and presentation, with a README.md at the top.
- **Secrets and backups:** API keys and database passwords stay in .env files that are not uploaded. Documents are also backed up in a shared drive.

6. Risks and How We Will Handle Them

Risk	What we will do
Unsafe food getting through	The AI only flags items. Doubtful food is rejected, and a person reviews every flagged item.
OCR misreading a date	Test on many different packages, let a person correct the date by hand, and flag results the OCR is unsure about.
AI service is slow, limited or down	Keep requests short, reuse repeated answers, and make sure the main features still work without the AI.
Poor Bangla answers from the chatbot	Test several prompts and limit answers to information from our own platform.
Someone falls behind	Clear roles, a look at everyone’s commits each week, and Week 9 kept free for catching up.
Too much for one semester	Build the must-have list first. Anything else waits or becomes future work.